Emergent heterogeneity in compartmental disease models and its relation to super-spreading

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1 Introduction

Real outbreaks have significant heterogeneity in the number of secondary cases – including *superspreading,* marked by a small fraction of individuals causing a disproportionately large fraction of realized transmission [11, 17]. Superspreading can help explain why epidemics break- and burn out in some times and places but not others, despite apparently similar contexts [10, 3]. Furthermore, control efforts incorporating superspreading into their design can be more effective [4, 13, 16].

Superspreading may be most salient for disease dynamics and control strategies at the time of outbreak [5]. For this reason, theoretical treatment of superspreading emphasizes individual-level transmission patterns, often while assuming a fully susceptible population [11, 8]. Theoretical analysis of superspreading has focused on effective, non-mechanistic models that link observed secondary case counts to a continuous random variable, the *individual reproductive number,* via a Poisson sampling process. Canonically, the individual reproduction number is modeled as a gamma distribution, giving rise to negative binomial secondary case distributions [11]. In this way, variation from the stochastic process of successful transmission is conceptually separated from a more intrinsic individual-level heterogeneity [in expected infectiousness] that may emerge from a plethora of potential differences in biology, contact patterns, or behaviors between individuals [2, 1, 6].

Over longer timescales, the processes of susceptible depletion and resupply, which are elegantly captured in mechanistic compartmental models, drive population-scale disease dynamics. Despite shared theory motivating models that highlight individual-level transmission events and superspreading, and those that emphasize population-level transmission dynamics, a gap persists in how each model framework treats individual-level transmission heterogeneity.

In its simplest form, the canonical ODE S[E]IR (Susceptible-Exposed-Infected-Removed) **[AA:** ***What's the harm of removing "Exposed"?*]** model assumes homogeneous rates of recovery and constant, uniform infectivity across all individuals during the infectious stage [7, 9]. Although single-infective-class compartmental models would seem ill-equipped to describe heterogeneity, even the SIR model yields exponentially distributed recovery times and thus, drive variation in the number of infections attributable to fractions of the population infected at a given time, while a changing effective reproduction number, caused by susceptible depletion, drives variation between fractions of the population infected at different times. In other words, even fully deterministic ODE models predict variation around basic reproductive number $\mathcal{R}_0$, the average number of secondary cases caused by a case in a fully susceptible population, in the expected secondary cases. We term this variation, *emergent heterogeneity.* In practice, epidemiological modelers also include *explicit heterogeneity* into their models, for example by considering age- or spatially-explicit compartments that may have different expected rates of mixing, transmission, or recovery [15]. Explicit heterogeneity reproduces superspreading in linear compartmental models [12].

We delineate the distinct sources of individual variation in transmission inscribed into a simple SIR model. We trace how these sources compose into observed or observable heterogeneity in different settings, for example contrasting how heterogeneity emerges in dynamical models versus linearized, or branching-process models. Throughout, we highlight the understudied but critically important role of disease dynamics *per se* in generating individual-level heterogeneity. We aim to spur the development of theory and methods that better integrate heterogeneity and dynamics, ultimately facilitating feedback between data on incidence and individual-level transmission chains, mechanistic model specification, identification, and validation, and intervention design.

Box 1. κ tutorial.

The mean and variance of a mixed Poisson distribution whose mixing distribution has mean μ and variance σ^2 are, respectively, μ and $\mu + \sigma^2$. Thus, both mixing and mixed Poisson distributions can be described using two quantities: mean μ and (generalized) *dispersion* parameter, κ, defined as σ^2/μ^2.

The statistic κ increases with variability around the mean and equals zero for a Dirac delta function, making it a desirable measure of dispersion.

The secondary-case distribution is a mixed Poisson distribution, with expected infectiousness as the mixing distribution. Hence, we use a single parameter, κ, to characterize dispersion in both secondary cases and expected infectiousness throughout this paper.

2 Results

Demographic stochasticity can generate emergent heterogeneity even in the absence of explicit differences in individual-level rates. In outbreaks modeled using the linearized SIR compartmental models (with susceptible, infected, and removed compartments), this heterogeneity can be quantitatively characterized.

In these models, infectious period follows an exponential distribution, resulting in an exponentially distributed expected infectiousness (or *individual reproductive number*) with $\kappa = 1$. Consequently, the secondary-case distribution (or *offspring distribution*), modeled as a Poisson mixture of the expected infectiousness, follows a geometric distribution with dispersion κ (see Box).

[JD: *Is that really what Box is doing, though? Or more about linking the emergent stochasticity in the deterministic vs. demographic-stochastic models?***]**

Panel (a) of Figure 1 shows patterns of emergent heterogeneity in secondary-case numbers during the early stage of an outbreak, when the population is fully susceptible. The distributions of both expected infectiousness and the number of secondary cases depend on $\mathcal{R}_0$, i.e., the average number of cases caused by a typical infector in a fully susceptible population.

Under a stochastic SIR model, where the number of secondary cases arises from Poisson draws from expected infectiousness, inequality in secondary cases–that is, the relationship between the fraction of new infections and the fraction of infectors–depends on the mean, whereas inequality in expected infectiousness does not (Figure 1, panel b). In fact, the share of Poisson noise in overall inequality shrinks as $\mathcal{R}_0$ increases, and the inequality in secondary cases approaches its theoretical limit, namely, the inequality in expected infectiousness (Figure 1, panel b). When $\mathcal{R}_0$ is small, an increasing fraction of infected individuals generate no secondary infections, so the individuals who do transmit make up a smaller fraction of all infectors. Indeed, the "20/80 rule"–that 80% of secondary infections arise from only 20% of infectious individuals–is attained when $\mathcal{R}_0$ falls just below 0.4. **[AA:** *don't know why***]** However, even when $\mathcal{R}_0$ exceeds the epidemic threshold, the distribution of secondary cases can still display substantial heterogeneity.

However, the distributions plotted in Figure 1 neglect the depletion of the susceptible population which occurs in real outbreak. To what extent does the incorporation of changes in the susceptible pool affect these distributions? To address this, we simulated simple, stochastic SIR epidemics with different $\mathcal{R}_0$ values in a population of 10000 individuals. For each infected individual, we recorded the number of secondary cases they generated during the outbreak. Panel (a) of Figure 2 shows realized distributions of secondary cases caused by individual infectors over the entire outbreaks. Unlike in the linearized scenario, where the susceptible pool is fully populated (Figure 1, panel a), the secondary case distributions remain indistinguishable across a broad range of the key parameter $\mathcal{R}_0$. This result is unexpected. Faster and larger dynamical changes occur when $\mathcal{R}_0$ is large (Figure S1), which would typically be expected to correspond to larger variation in the expected infectiousness distribution and, hence, secondary-case distribution.

To clarify these counterintuitive observations, we considered a standardized deterministic SIR model in which the time is measured in units of the mean infectious period. We examined how heterogeneity evolves in epidemiologically relevant cohorts, defined as groups infected simultaneously, over the course of the outbreak. A large $\mathcal{R}_0$ initially generates significant expected infectiousness. Later, rapid depletion of susceptible individuals reduces the expected infectious-

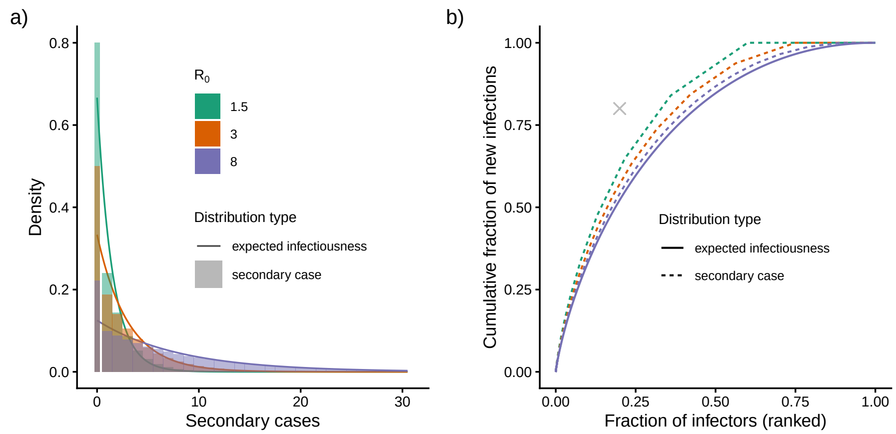

Figure 1: **Heterogeneity emerges even from a simple, linearized compartmental model** due to implicit variation in recovery times among infectors. Panel a: Both expected infectiousness (density curves) and secondary case distributions (density histograms) for the outset of an SIR epidemic, where the population is fully susceptible depend on $\mathcal{R}_0$. Panel b: Inequality curves, defined as the cumulative fraction of new infections versus the cumulative fraction of infectors, computed from the expected infectiousness across varying $\mathcal{R}_0$, are identical and visually indistinguishable due to overplotting. In contrast, inequality curves derived from the secondary case distribution exhibit differences. As $\mathcal{R}_0$ increases, the inequality observed in the secondary case distribution decreases and approaches the expected infectiousness inequality curve. The cross marks "20-80" point, where 20% of infectors cause 80% of infections. In panel a, because the first bin (at zero) sits at the boundary of support for each distribution, we have plotted this bin as double the density and half the width; this adjustment preserves area-to-area correspondence with the probability density function, while facilitating visual comparison of the heights of the density and mass functions.

ness. In contrast, a small $\mathcal{R}_0$ initially causes smaller expected infectiousness, but the slower susceptible exhaustion allows for considerable expected infectiousness (Figure S2). We then split the total variance in expected infectiousness over the course of the outbreak into between- and within-cohort components. We found that epidemics with larger $\mathcal{R}_0$ have larger between-cohort variation, as expected. However, this is balanced by smaller within-cohort variation (Figure 2, panel b). We can in fact prove that the mean and variance of expected infectiousness, conditioned on the entire outbreak, is equal to one, regardless of the value of $\mathcal{R}_0$ (Section S.1). Thus, the dispersion κ is also one.

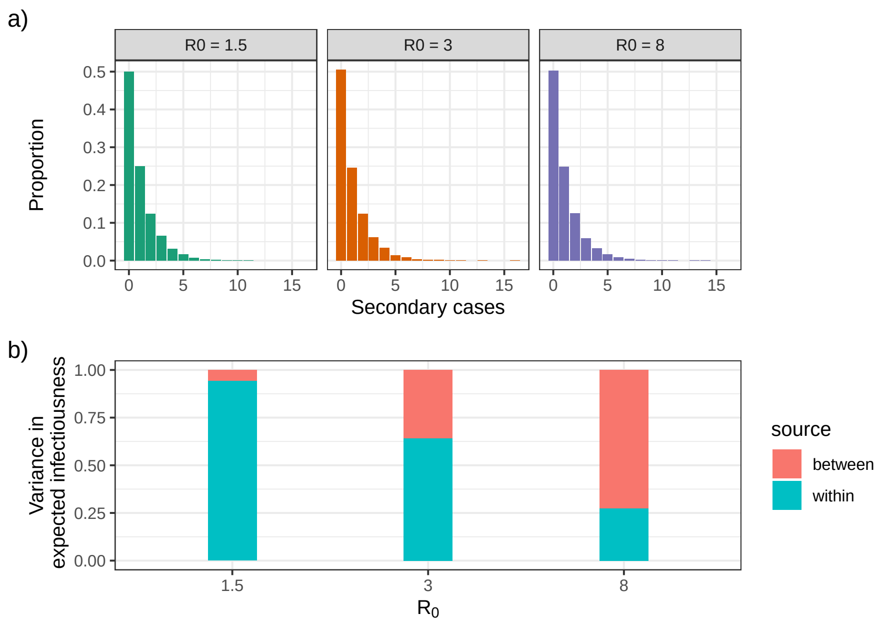

Figure 2: **Indistinguishable empirical secondary-case distributions for epidemics of different strengths** conditioned over the outbreak with $\kappa = 1$. Panel a: Epidemic outbreaks were simulated for a population of 10000, and the number of secondary cases per infector was recorded at the time of outbreak extinction. Panel b: Identical variance in expected infectiousness for epidemics of different strengths modeled using standardized compartmental models. As $\mathcal{R}_0$ increases, between-cohort variance rises, and within-cohort variance falls, maintaining constant total variance of one.

We then investigated how expected infectiousness would evolve over time? To do so, for different values of $\mathcal{R}_0$, we simulated an outbreak using the standardized deterministic SIR model and evaluated it at several cutoff (time) points. At each cutoff point, we calculated heterogeneity in expected infectiousness between cases generated by that time. In the early stage of the outbreak, observed heterogeneity was lower than in the linearized model or the full outbreak, and it varied by $\mathcal{R}_0$. From the beginning, in epidemics with large $\mathcal{R}_0$, the between-cohort variation was the main driver of heterogeneity. By the end of the outbreak, heterogeneity in expected infectiousness among different $\mathcal{R}_0$ values became indistinguishable (Figure 3).

We also examined an idealized scenario in which, for each cutoff point, we computed heterogeneity in expected infectiousness over the entire outbreak, but considering those cohorts that had become infectious up to that time. In epidemics with lower $\mathcal{R}_0$, the mean and standard deviation of expected infectiousness for early-infected cohorts almost matched those from the linearized model, i.e., mean and standard deviation equal $\mathcal{R}_0$ (Figure 4). Including cohorts infected later, but before the outbreak peak, reduced variability as measured by κ. Including cohorts infected after the peak increased variability. By the end of the outbreak, all epidemics with different $\mathcal{R}_0$ agreed on the heterogeneity in expected infectiousness, measured by κ. Regardless of $\mathcal{R}_0$, within-cohort variation primarily drove heterogeneity among early-infected cohorts. However, after the peak, the susceptible depletion reduced the share of within-cohort variation, a reduction that was more pronounced in epidemics with large $\mathcal{R}_0$.

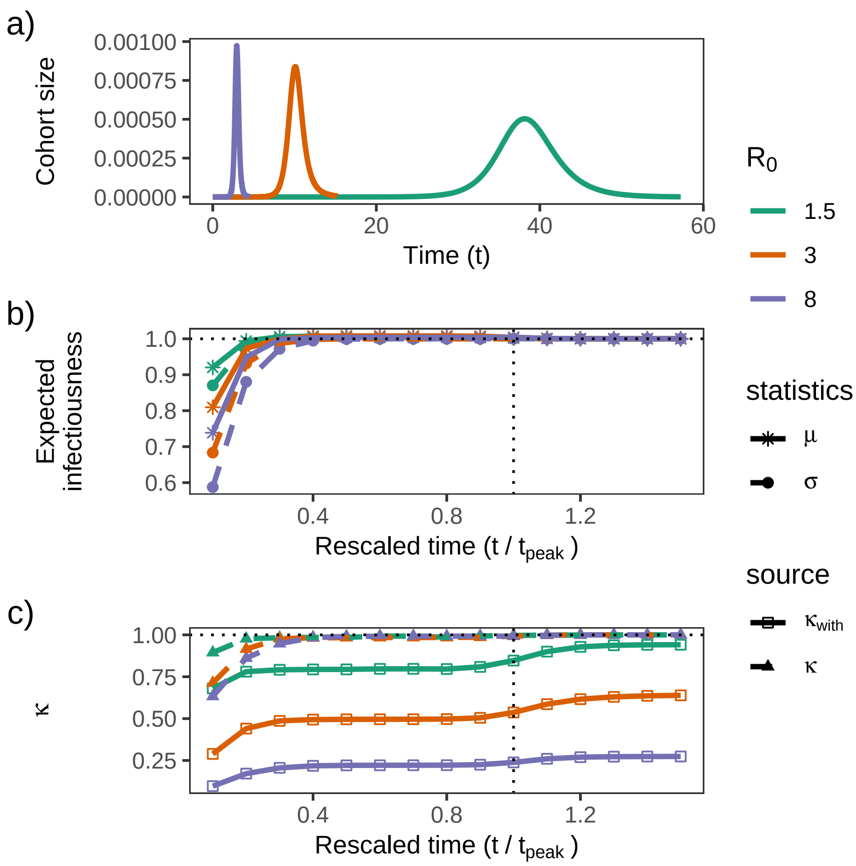

Figure 3: **The expected infectiousness evolves over the course of the outbreak.** Panel a: The size of the cohort infected during $[t, t + \Delta t]$ equals the area under the incidence curve on that interval. Time is in units of the mean infectious period, and $\Delta t = 2 \times 10^{-4} t_{\text{peak}}$ (where t_{peak} is the time of peak incidence). Panel b: The mean and standard deviation of expected infectiousness at each time are computed using the realized cases up to that point. In stronger epidemics, t_{peak} is smaller, and almost all infected individuals are still infectious well before t_{peak}, leading to lower heterogeneity compared to epidemics with smaller $\mathcal{R}_0$. Panel c: κ climbs over the course of the outbreak. Epidemics with small $\mathcal{R}_0$ take longer to reach their peak. Even fractions of t_{peak} are long enough to reveal differences in infectious period among cohort members, resulting in a relatively large within component of κ (measured as the ratio of within-cohort variance to the squared mean). After the peak, the difference in infectious period among individuals in the largest cohort, who infected at the peak time, drives a slight increment in within-cohort variation. For panels b and c, the simulation was stopped at each time point to compute the y-axis value. The y-axis value at each time point was computed by assigning to each cohort the realized case reproductive number up to that time. In panels b and c, time units measured in the mean infectious period are again scaled relative to the peak time.

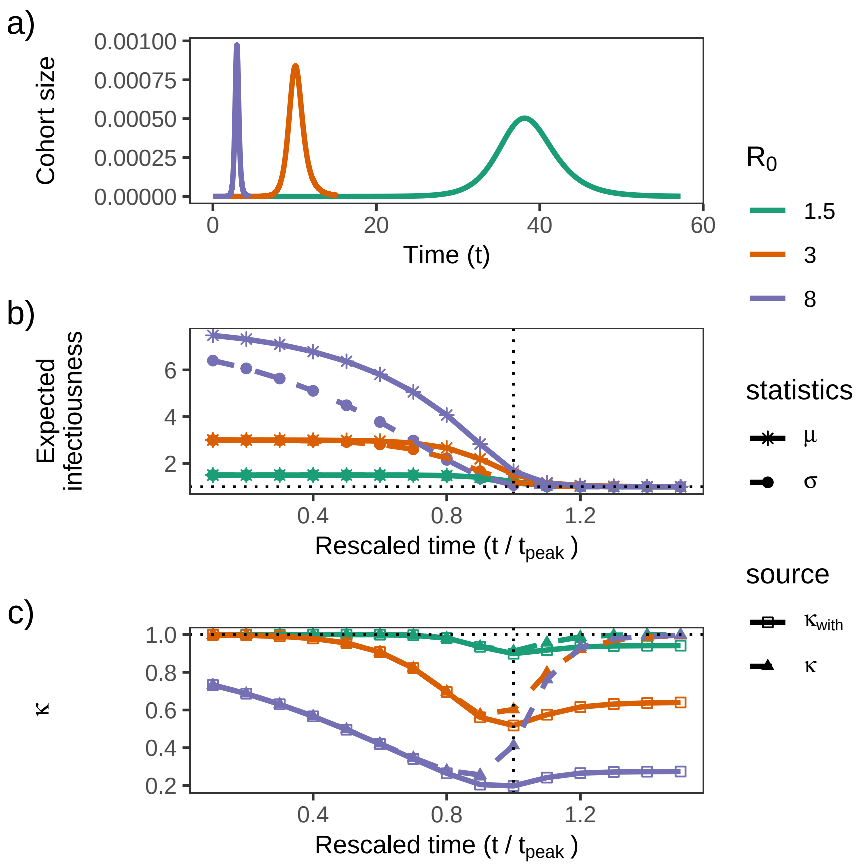

Figure 4: **Forecasting the distribution of expected infectiousness evolves as the outbreak unfolds.** Panel a: Cohort size infected during $[t, t + \Delta t]$ is measured as the area under the incidence curve over this interval ($\Delta t = 2 \times 10^{-4} t_{\text{peak}}$, where t_{peak} is the time of peak incidence). Panel b: At each time point, the mean and standard deviation of expected infectiousness are calculated using only cohorts infected up to that specific time, incorporating the subsequent contributions of these cohorts to disease transmission. By "rescaled," we mean that time has first been scaled by the mean infectious period and is then measured relative to the outbreak peak. The mean of expected infectiousness is larger for early-infected cohorts and depends on the key parameter $\mathcal{R}_0$. As time goes by, later-infected cohorts, who have larger sizes, experience susceptible pool depletion, decreasing expected infectiousness. After the peak, the susceptible population size stabilizes, and the mean and standard deviation approach one. Panel c: The within-cohort component of the squared coefficient of variation κ_{with}, measured as the ratio of the within-cohort variance to the squared mean, is the main driver of heterogeneity among the early-infected cohorts, as they experience a similar size of susceptible population. As the outbreak unfolds, the decline in the susceptible population drives between-cohort variation, which is more pronounced in epidemics with large $\mathcal{R}_0$. κ reaches its minimum around the outbreak peak; large cohorts experience sharp depletion of susceptibles, and the impact of variation in recovery time fades, declining the variance, and hence κ. After the peak, κ rebounds to one. For panels b and c, the y-axis value at each time point was computed using only the cohorts that had been infected up to that time.

3 Discussion

Quantifying heterogeneity in secondary-case distributions helps better understand how epidemics unfold and how they can be contained. Although there have been efforts to both quantitatively describe individual-level heterogeneity in contact rates [17] and to parameterize biologically realistic secondary case distributions, abstracted from the mechanism [11], these efforts have been largely separate from the construction and evaluation of the compartmental models that enable epidemic forecasting and intervention design (but see [12]). Here, we expand the bridge between individual- and population perspectives of the heterogeneity by studying epidemiologically-relevant cohorts, those infected at the same time.

To do so, we first describe a statistic, κ, for quantifying the emergent heterogeneity entailed in the SIR models and their stochastic realizations. Next, we compare the emergent heterogeneity from deterministic ODE SIR models to the dispersion in secondary case distributions resulted from their stochastic realizations. Finally, we consider how different time frames dictate different views of emergent heterogeneity, and how expected infectiousness evolves through time in epidemics of different strengths.

We observed that early in the outbreak, mean and variance of the expected infectiousness vary by $\mathcal{R}_0$, so do those of secondary-case distributions. However, when conditioned over the entire outbreak, they become independent of $\mathcal{R}_0$. We indeed prove that, provided a negligible fraction of initial infectors, both mean and variance are equal to one, regardless of $\mathcal{R}_0$. Disentangling the within- and between-cohort variances reveals that in epidemics with large $\mathcal{R}_0$, unlike those with smaller $\mathcal{R}_0$, the between-cohort variance dominates the within-cohort variance.

In real-time analyses, κ, used as a measure of heterogeneity, remains below one.

There are some limitations to our work. The underlying model is based on a closed, well-mixed population. It is not clear how heterogeneity in mixing propensity might impact the independence of heterogeneity in expected infectiousness from $\mathcal{R}_0$ and, hence, alter κ. Working with SIR models implicitly implies an exponential infectious period. Investigating the impact of infectious period distribution on the heterogeneity in expected infectiousness is left for future work.

[TG: *Can we make a note about for epidemics with large R0, if you don't start tracking cases right from the beginning, you'll already underestimate cases/case* **]** **[JD:** *Yes, this should go into the paper.* **]**

In SIR-type epidemics, the effective reproduction number declines as the susceptible population becomes depleted. Consequently, if surveillance begins after the initial transmission generations, estimates of secondary-cases will represent the reduced effective reproduction number rather than $\mathcal{R}_0$[14]. As a result, delayed surveillance may yield substantially lower estimates, especially when $\mathcal{R}_0$ is large, due to rapid depletion of the susceptible pool.

4 Materials and Methods

The values of secondary cases in panel (a) of Figure 1 were generated by computing the geometric probability mass function with mean $\mathcal{R}_0$ at points $0, 1, \ldots, 30$. As for the expected infectiousness, we evaluated the exponential probability density function with mean $\mathcal{R}_0$ at 300 equally distanced point in the interval $[0, 30]$.

To generate panel (b) of Figure 1, we followed the Lloyd-Smith's approach and computed the inequality in the activity distribution [11, SI, Sec. 2.2.5]. More specifically, we used the relation $F_{\text{trans}}(x) = \frac{1}{\mathcal{R}_0^2} \int_0^x ve^{-\frac{v}{\mathcal{R}_0}} dv$ which is the fraction of cases due to those caused up to x secondary cases. The fraction of cases due to those caused more than x cases would be $1 - F_{\text{trans}}(x)$, which is equal to $(1 + \frac{x}{\mathcal{R}_0})e^{-\frac{x}{\mathcal{R}_0}}$. The population fraction of the individuals infected more than x cases is $e^{-\frac{x}{\mathcal{R}_0}}$. We used a similar approach to compute the inequality in the secondary case distribution: We used the relation $F_{\text{trans}}(x) = \frac{1}{\mathcal{R}_0} \sum_{v=0}^{x} vG(v, \mathcal{R}_0)$ which is the fraction of cases due to those caused up to x secondary cases. Here $G(v, \mathcal{R}_0)$ is a geometric distribution with mean $\mathcal{R}_0$. The fraction of cases due to those caused more than x cases equals $1 - F_{\text{trans}}(x)$. The population fraction of these individuals would be $1 - P(x)$, where $P(x)$ is the cumulative distribution function of a geometric distribution with mean $\mathcal{R}_0$ evaluated at x.

To generate panel (a) of Figure 2, we used the individual based simulation and computed the number of secondary cases in a population of size 10^4. The simulation was initiated with one case.

We used a simple deterministic SIR model to compute the mean and variance of expected infectiousness $\mathcal{R}_{\text{c}}$. First, we scaled time by the mean infectious period, so the resulting SIR model then depends on one parameter: $\mathcal{R}_0$. The SIR differential equations were numerically solved for the proportions of susceptible $x(t)$ and infectious $y(t)$ at each time point t. The time interval for integration was set to $[0, 100]$, well after the outbreak died out. We then used the **R** function `approxfun` to construct a function, $X(t)$, that linearly interpolates the time-series $(t, \mathcal{R}_0 x(t))$. We associated a cohort to each of the first 60% time points in the time-series t (We used a 60% cutoff to ensure cohorts had almost recovered by the end of the simulation period) and calculated the cohort-specific mean and variance. More specifically, for the cohort infected at time point τ, the mean, $\mu(\tau)$, and variance $\sigma^2(\tau)$ of $\mathcal{R}_{\text{c}}$ read as:

$$\mu(\tau) = \int_{t>\tau} f(t-\tau) \int_{\tau}^{t} \mathcal{R}_0 x(s) ds \, dt \,, \tag{1a}$$

$$\mathbb{E}[\mathcal{R}_{\text{c}}^2(\tau)] = \int_{t>\tau} f(t-\tau) \left(\int_{\tau}^{t} \mathcal{R}_0 x(s) ds \right)^2 dt \,, \tag{1b}$$

$$\sigma^2(\tau) = \mathbb{E}[\mathcal{R}_{\text{c}}^2(\tau)] - \mu^2(\tau). \tag{1c}$$

To calculate these quantities for the cohort infected at τ, the interpolating function $X(t)$ was integrated from τ to t, which is the case reproductive potential associated with the cohort fraction recovered at time point t. The mean $\mathcal{R}_{\text{c}}$ for each cohort $\mu(\tau)$ was then the integral of the case reproductive potential weighted by the infectious period density function. We used the same approach to compute $\mathbb{E}[\mathcal{R}_{\mathrm{c}}^2(\tau)]$ and variance of $\mathcal{R}_{\mathrm{c}}$ for each cohort ($\sigma^2(\tau)$). In Figure S2, the middle panel (panel b) was generated using this approach.

We obtained the mean of expected infectiousness $\mathcal{R}_{\mathrm{c}}$ by integrating the cohort-specific mean $\mu(\tau)$ against the incidence, $i(\tau) = \mathcal{R}_0 x(\tau) y(\tau)$, and normalizing the result by the final size, $\int i(t)dt$. To compute the between cohort variance, we integrated the squared cohort mean $\mu^2(\tau)$ weighted by the incidence $i(\tau)$ and divided it by the final size. The result minus the squared mean of $\mathcal{R}_{\mathrm{c}}$ equaled the between cohort variance. The within-cohort variance was computed by taking integral over the cohorts' variance $\sigma^2(\tau)$ weighted by the incidence. The result was also normalized by the final size. The total variance was the sum of the between- and within-variances. We also calculated the total variance independently; we integrated $\mathbb{E}[\mathcal{R}_{\mathrm{c}}^2(\tau)]$ weighted by the incidence and divided it by the final size. The result minus the squared mean of $\mathcal{R}_{\mathrm{c}}$ yielded the total variance. Both approaches produced the same value for the total variance.

We used deterministic simulations to compute the y-axis values in the panels b and c of Figure 3. Specifically, for each time point t_f, we considered the cohorts infected up to time t_f. We then computed cohort statistics by calculating the cases generated by these cohorts up to time t_f. For example, for a cohort infected at time point $\tau < t_f$, the mean, $\mu(\tau, t_f)$, and variance of $\mathcal{R}_{\mathrm{c}}$, $\sigma^2(\tau, t_f)$, read as:

$$\mu(\tau, t_f) = \int_{t=\tau}^{t_f} f(t-\tau)dt \int_{\tau}^{t} \mathcal{R}_0 x(s)ds + F(t_f - \tau)\int_{\tau}^{t_f} \mathcal{R}_0 x(s)ds, \quad \text{(2a)}$$

$$\begin{aligned}\mathbb{E}[\mathcal{R}_{\mathrm{c}}^2(\tau, t_f)] = &\int_{t=\tau}^{t_f} f(t-\tau)dt \left(\int_{\tau}^{t} \mathcal{R}_0 x(s)ds\right)^2 \\ &+ F(t_f-\tau)\left(\int_{\tau}^{t_f} \mathcal{R}_0 x(s)ds\right)^2, \quad \text{(2b)}\end{aligned}$$

$$\sigma^2(\tau, t_f) = \mathbb{E}[\mathcal{R}_{\mathrm{c}}^2(\tau, t_f)] - \mu^2(\tau, t_f), \quad \text{(2c)}$$

where $F(\cdot)$ is the survival function associated with the infectious period density $f(\cdot)$. The mean and variance of $\mathcal{R}_{\mathrm{c}}$ are then

$$\mu(t_f) = \frac{\int_0^{t_f} i(\tau)\mu(\tau, t_f)d\tau}{\int_0^{t_f} i(t)dt}, \quad \text{(3a)}$$

$$\sigma^2(t_f) = \frac{\int_0^{t_f} i(\tau)\mathbb{E}[\mathcal{R}_{\mathrm{c}}^2(\tau, t_f)]d\tau}{\int_0^{t_f} i(t)dt} - \mu^2(t_f). \quad \text{(3b)}$$

The between-cohort component of κ at time point t_f is $\kappa_{\mathrm{bet}}(t_f) = \sigma^2_{\mathrm{bet}}(t_f)/\mu^2(t_f)$, where

$$\sigma^2_{\mathrm{bet}}(t_f) = \frac{\int_0^{t_f} i(\tau)\mu(\tau, t_f)^2}{\int_0^{t_f} i(t)dt} - \mu^2(t_f).$$

o generate panels (b) and (c) of Figure 4, as with the corresponding panels of Figure 3, at each time point t_f, we considered only cohorts infected by t_f. Instead, we counted the total cases caused by these cohorts during the outbreak. To do so, we simply substituted $\mu(\tau)$ and $\mathbb{E}[\mathcal{R}_c^2(\tau)]$ for $\mu(\tau, t_f)$ and $\mathbb{E}[\mathcal{R}_c^2(\tau, t_f)]$ in Equation (3), where $\mu(\tau)$ and $\mathbb{E}[\mathcal{R}_c^2(\tau)]$ were provided in Equation (1).

All integrations were done using `lsoda` method with the **R** package `deSolve`. All simulations were carried out using **R** `4.5.2`. Code for all numerical simulations is housed at: `https://github.com/dushoff/kappaCode`. We solved all integrals across a range of values for $\mathcal{R}_0$, using the starting values $y_0 = 10^{-9}; x_0 = 1 - 10^{-9}$ to represent the limiting case in which there are no exogenous cases. In building these simulations, we used a range of time step sizes, noting convergence towards known and conjectured values (e.g., epidemic final size, mean case reproduction number, variance in case reproduction number) as resolution increased.

S.1 Analytical Results

S.1.1 Preliminaries

The standard SIR model is described by the following equations:

$$\dot{x}(t) = -\mathcal{R}_0 x(t) y(t),$$
$$\dot{y}(t) = \mathcal{R}_0 x(t) y(t) - y(t),$$

where $x(t)$ and $y(t)$ denote the proportion of susceptible and infectious individuals, respectively. Time has been normalized by the recovery rate. Let $i(t)$ denote the incidence $\mathcal{R}_0 x(t) y(t)$, and define the size of the epidemic as follows:

$$Z = \int_0^\infty dt i(t).$$

Let f denote the distribution of residence times in the infectious compartment. Define $F(t)$ as the corresponding survival distribution, that is

$$F(t) = 1 - \int_0^t d\tau f(\tau) = \int_t^\infty d\tau f(\tau). \tag{S.1}$$

Note that $F(t)$ is the probability that an individual infected at time 0 remains infectious after time t.

We define the *expected case reproductive number* between time τ and ρ as

$$C(\tau, \rho) = \mathcal{R}_0 \int_\tau^\rho dt\, x(t).$$

The k^{th} weighted raw moment of C is then defined as

$$C_k = \int_0^\infty d\tau \int_\tau^\infty d\rho\, w(\tau, \rho)(C(\tau, \rho))^k, \tag{S.2}$$

where $w(\tau, \rho) = i(\tau) f(\rho - \tau)$ is the size of the cohort that became infectious at time τ and recovered at time ρ.

S.1.2 The Statement

The excess variation of the individual case reproductive number in the standard SIR model outbreak is 1. Mathematically, this amounts to proving that

$$\boxed{\kappa = \frac{C_0 C_2}{C_1^2} - 1 = 1.} \tag{S.3}$$

S.1.3 Proof

We will calculate C_0, C_1 and C_2. We can calculate C_0 and C_1 without making any assumptions about the functional form of f (we don't have to assume that the wait times are exponentially distributed). We do need F to be memoryless, i.e., obey $F(a+b) = F(a)F(b)$, to get the "right" value of C_2. Calculating each of these moments essentially involves multiple manipulations of the limits of the integrals involved.

From Equation S.2, we have

$$\begin{aligned} C_0 &= \int_0^\infty d\tau \int_\tau^\infty d\rho \; w(\tau,\rho) \\ &= \int_0^\infty d\tau \int_\tau^\infty d\rho \; i(\tau) f(\rho - \tau) \end{aligned}$$

Substituting $\omega + \tau$ for ρ in the inner integral, results in

$$C_0 = \int_0^\infty d\tau \; i(\tau) \int_0^\infty d\omega \; f(\omega).$$

Since f is the residence time distribution (note we're not invoking the fact that f is an exponential distribution!), the inner integral is simply 1. It follows that

$$\begin{aligned} C_0 &= \int_0^\infty d\tau \; i(\tau) \\ &= Z. \end{aligned}$$

The next step is to prove that $C_1 = Z$. From Equation S.2, we have

$$\begin{aligned} C_1 &= \int_0^\infty d\tau \int_\tau^\infty d\rho \; w(\tau,\rho) C(\tau,\rho) \\ &= \mathcal{R}_0 \int_0^\infty d\tau \int_\tau^\infty d\rho i(\tau) f(\rho - \tau) \int_\tau^\rho dt x(t). \end{aligned}$$

Instead of first integrating over t from τ to ρ and then over ρ from τ to ∞, we can first integrate over ρ from t to ∞ and then integrate over t from τ to ∞, i.e.,

$$C_1 = \mathcal{R}_0 \int_0^\infty d\tau \int_\tau^\infty dt i(\tau) x(t) \int_t^\infty d\rho f(\rho - \tau).$$

Substitute $\rho = \omega + \tau$ in the innermost integral and using Equation S.1, we get

$$\begin{aligned} C_1 &= \mathcal{R}_0 \int_0^\infty d\tau \int_\tau^\infty dt\ i(\tau)x(t) \int_{t-\tau}^\infty d\omega f(\omega) \\ &= \mathcal{R}_0 \int_0^\infty d\tau \int_\tau^\infty dt\ i(\tau)x(t)F(t-\tau). \end{aligned}$$

Note again that we have not used the fact that f is exponential, we have merely used the definition of F as an integral of f. By interchanging the order of the integrals once more, we have

$$\begin{aligned} C_1 &= \mathcal{R}_0 \int_0^\infty d\tau \int_\tau^\infty dt\ i(\tau)x(t) \int_{t-\tau}^\infty d\omega f(\omega) \\ &= \mathcal{R}_0 \int_0^\infty dt\ x(t) \int_0^t d\tau\ i(\tau)F(t-\tau). \end{aligned}$$

Note that the inner integrand counts the number of individuals that entered the infectious state at some time $\tau < t$ and remain infectious at time t. When we integrate over all times $\tau < t$, we simply get the number of individuals who are infections at time t, i.e., $y(t)$. So,

$$C_1 = \mathcal{R}_0 \int_0^\infty dt\ x(t) \int_0^t d\tau\ i(\tau)F(t-\tau) \tag{S.4}$$

$$= \mathcal{R}_0 \int_0^\infty dt\ x(t)y(t) \tag{S.5}$$

$$= \int_0^\infty dt\ i(t) \tag{S.6}$$

$$= Z. \tag{S.7}$$

The next step is to show $C_2 = 2Z$. Given Equation S.2, we have

$$\begin{aligned} C_2 &= \int_0^\infty d\tau \int_\tau^\infty d\rho\ w(\tau,\rho) \left(C(\tau,\rho)\right)^2 \\ &= \mathcal{R}_0^2 \int_0^\infty d\tau \int_\tau^\infty d\rho\ i(\tau)f(\rho-\tau) \left(\int_\tau^\rho dt x(t)\right)^2 \end{aligned}$$

Now we break down the square integral into a double integral

$$\begin{aligned} \left(\int_\tau^\rho x(t)dt\right)^2 &= \left(\int_\tau^\rho x(t)dt\right)\left(\int_\tau^\rho x(s)ds\right) \\ &= \left(\int_\tau^\rho x(t)dt\right)\left(\int_\tau^\rho x(s)ds\right) \\ &= \int_\tau^\rho \int_\tau^\rho dtds\ x(t)x(s). \end{aligned}$$

In $t-s$ space, this is integrating $x(t)x(s)$ over a square region whose vertices are (τ,τ), (τ,ρ), (ρ,τ) and (ρ,ρ). This is twice the integral over the triangle whose vertices are (τ,τ), (τ,ρ) and (ρ,τ). So,

$$\left(\int_\tau^\rho x(t)dt\right)^2 = 2\int_\tau^\rho ds \int_\tau^s dt\ x(t)x(s)$$

Plugging this back into the expression for C_2 yields

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty d\tau \int_\tau^\infty d\rho \int_\tau^\rho ds \int_\tau^s dt\ i(\tau)f(\rho-\tau)x(t)x(s).$$

First, we reorder the integrals over ρ and s and change the limits appropriately

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty d\tau\ i(\tau) \int_\tau^\infty ds\ x(s) \int_s^\infty d\rho \int_\tau^s dt f(\rho-\tau)x(t).$$

Next, we reorder the integrals over ρ and t to get

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty d\tau\ i(\tau) \int_\tau^\infty ds\ x(s) \int_\tau^s dt\ x(t) \int_s^\infty d\rho f(\rho-\tau).$$

By substituting $\rho=\omega+\tau$ and using Equation S.1, we get

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty d\tau\ i(\tau) \int_\tau^\infty ds\ x(s) \int_\tau^s dt\ x(t)F(s-\tau).$$

We then reorder the integrals over τ and s

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty ds\ x(s) \int_0^s d\tau \int_\tau^s dt\ x(t)i(\tau)F(s-\tau).$$

Swapping the integrals over τ and t, results in

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty ds\ x(s) \int_0^s dt\ x(t) \int_0^t d\tau\ i(\tau)F(s-\tau).$$

Now, we use the fact that the survival function F is memoryless, i.e., $F(a+b) = F(a)F(b)$, and replace $F(s-\tau)$ with $F(s-t)F(t-\tau)$

$$C_2 = 2\mathcal{R}_0^2 \int_0^\infty ds\ x(s) \int_0^s dt\ x(t) \int_0^t d\tau\ i(\tau)F(s-t)F(t-\tau).$$

Next, we use the same argument we used to go from Equation S.4 to Equation S.5, which results in

$$\begin{aligned}
C_2 &= 2\mathcal{R}_0^2 \int_0^\infty ds\ x(s) \int_0^s dt\ x(t)F(s-t)y(t) \\
&= 2\mathcal{R}_0 \int_0^\infty ds\ x(s) \int_0^s dt\ i(t)F(s-t) \\
&= 2\mathcal{R}_0 \int_0^\infty ds\ x(s)y(s) \\
&= 2\int_0^\infty ds\ i(s) \\
&= 2Z.
\end{aligned}$$

Finally, replacing C_0, C_1, and C_2 by Z, Z, and $2Z$, respectively, in Equation S.3, results $\kappa = 1$.

Supplementary Figures

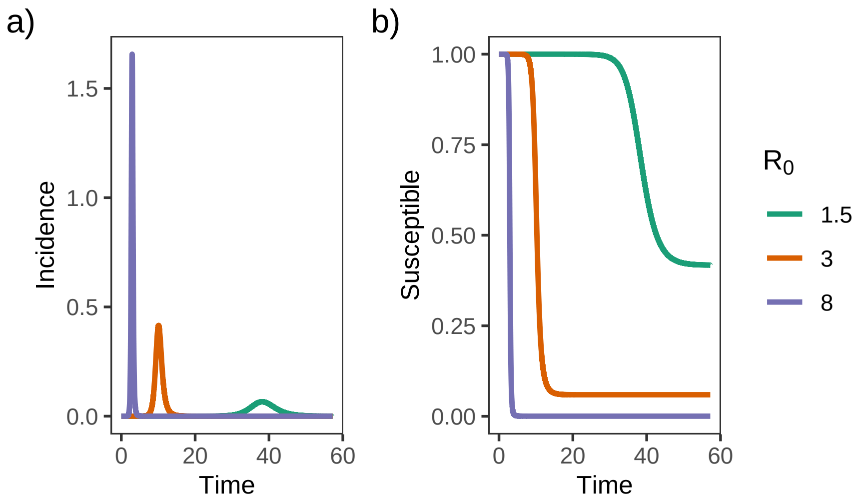

Figure S1: **The susceptible population changes during an outbreak.** Outbreaks with higher $\mathcal{R}_0$ values lead to greater variation in incidence and more rapid depletion of the susceptible pool. We modeled these outbreaks using a deterministic compartmental approach where time is in units of mean infectious period.

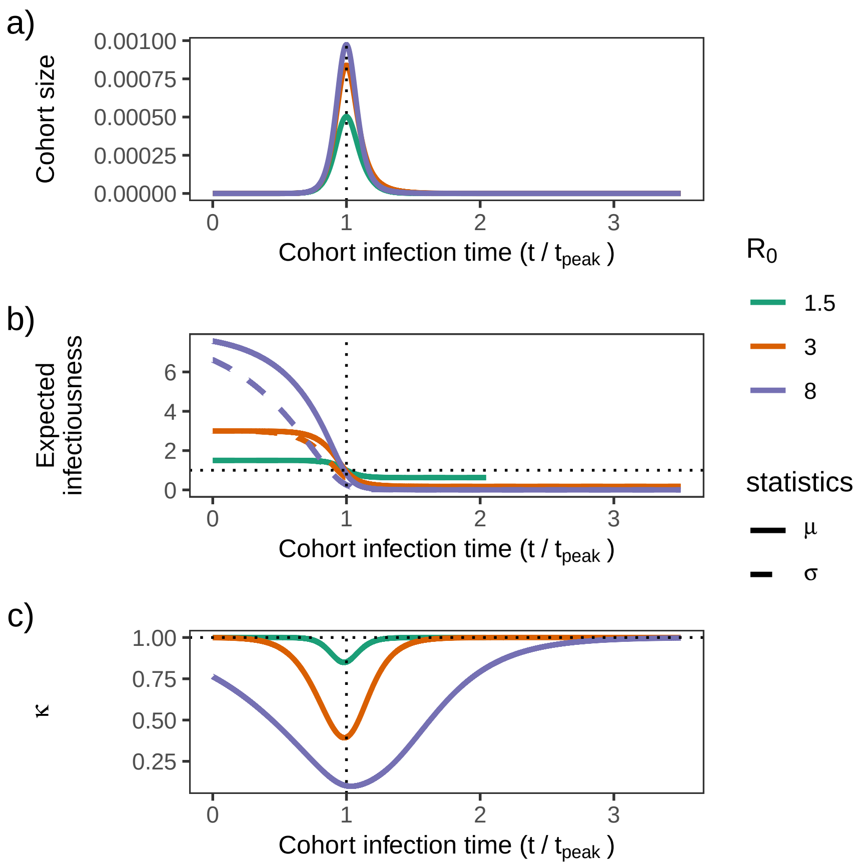

Figure S2: **Expected infectiousness of cohorts.** Panel a depicts cohort size as a function of the time of infection, in units of the mean infectious period divided by the incidence peak time. Individuals infected during $[t, t + \Delta t]$ form a cohort, with $\Delta t = 2 \times 10^{-4}$. In Panel b, the mean and standard deviation of expected infectiousness associated with each cohort are plotted. Early in the outbreak, the susceptible pool shrinks slowly, and differences in recovery time mainly cause variations in case numbers. As a result, the average for cases is about the same as in a simple linearized model. Near the outbreak's peak, faster susceptible loss partly compensates for the inequality stemming from the infectious period, and hence, the squared coefficient of variation κ decreases (Panel c). After the peak, once the susceptible pool has become nearly constant again, variation in the infectious period once more drives variation within cohorts; κ approaches one. In stronger outbreaks (higher $\mathcal{R}_0$), it takes longer (in rescaled time units) for the susceptible population to stabilize, explaining the slower move of κ toward one.

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